

AISEE-BIN: Flower Dome progress

Status update for the team · 1 October 2026 · Gardens by the Bay, Flower Dome POC



In short: on 1 October we tested phone navigation on site at the Flower Dome.

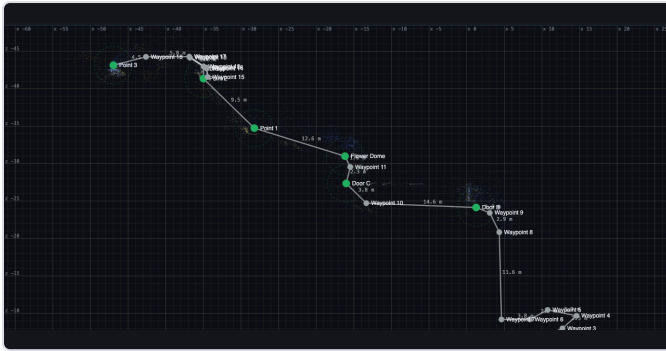
- **Inside the dome it worked.** On a route our tester scanned with the iPhone app, the phone guided her to Points 1, 2 and 3. She heard "Point X ahead" just before each one.
- **Around the entrance it struggled.** At Points A, B and C the phone had trouble working out where it was.
- **An Immersal scan fixed the entrance.** Using last week's Immersal scan of the entrance, the phone found itself immediately and knew roughly where it was.
- **Joining several Immersal scans into one big map isn't reliable yet.** That is the main problem left to solve.

What we have built

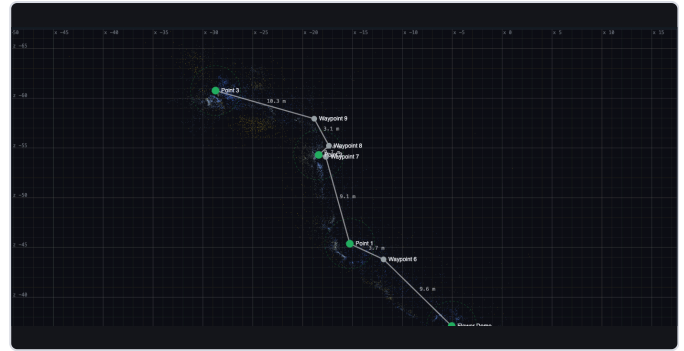
Piece	What it does now
iPhone app TestFlight, build 10	• Navigate: spoken turn-by-turn guidance with a live map, plus announcements for nearby exhibits and hazards. • Author: scan a place once, mark its stops and publish it. • Field Test: on-site tools to link scans, mark points, check accuracy and send a log. • Positioning: uses Immersal either on the phone or via the Immersal server, chosen with a switch.
Android app GitHub release 0.1.1	Glasses-only version for phones without ARCore (tested on a Huawei Nova 5T). It shows smooth video from the AiSee glasses, uses Immersal to position, and has a live map and spoken announcements.
Web map editor aiseebin.flowsxr.com	• Shows each scanned map as a point cloud, and keeps every saved version. • Lets you add stops, walkways, spoken descriptions and an announce radius for each place. • Lets you line up several Immersal scans into one map. • The apps pick up the latest version on their next launch.
Shared backend	• Maps, versions and Field Test results are stored on our server. • Testers' logs also arrive there, so we can see what happened on site without being there.

What we tested on 1 October

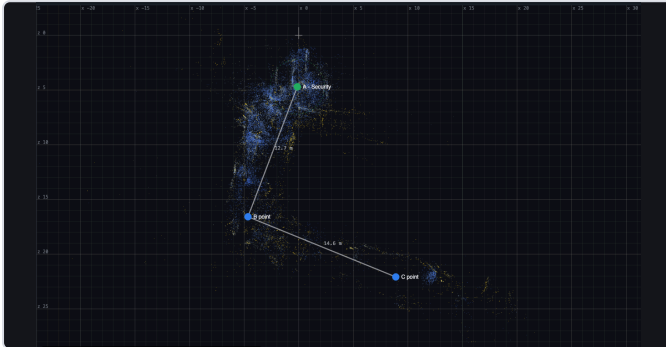
Test	What happened	Result
Five Immersal scans as one map last week's scans, lined up by hand	Our tester marked the tour points and then checked them. Linking the scans on the walk kept timing out on mobile data.	Not usable yet. The checks were 13–47 m off. A walk that matched all four scans gave new positions, but they still disagree with the marked points.
Phone scan "Flower Dome 1 Oct" Author mode, about 108 m, Point A to Point 3	Published on site after a couple of failed uploads. She then navigated to Points 1, 2 and 3 and back, and recorded her screen.	Worked well inside the dome. Each announcement came just before its point. Weak at A, B and C near the entrance.
Phone scan "Flower Dome 1 Oct map 2" the actual priority-lane route	Retested a little later as it was getting darker.	Stuck on "Relocalizing" at the entrance. The barrier graphics and the change in light are likely causes.
Immersal entrance scan "Entrance Test Oct 1st" last week's scan; A, B and C placed by eye in the editor	Loaded on the phone and tested at the entrance.	Found its position straight away and knew roughly where it was. The points themselves are rough, because they were placed by eye.



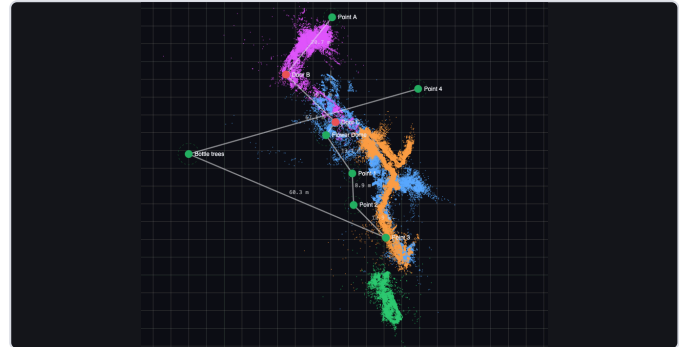
"Flower Dome 1 Oct": the phone scan from Point A through Doors B and C to Points 1, 2 and 3, as published from the app. The grey points are waypoints recorded on the walk.



"Flower Dome 1 Oct map 2": the priority-lane route, from Flower Dome to Point 3. On the retest it got stuck relocalizing at the entrance.



"Entrance Test Oct 1st": last week's Immersal entrance scan with A, B and C added in the editor. Navigation found its position immediately.



The five Immersal scans (one colour each), lined up by hand, with today's marked points. They look roughly right, but on site they were 13–47 m off.

What we learned

- **Phone scans (ARKit) work well where there is texture and steady light.** Inside the dome that was enough to guide someone. At the entrance the phone often failed to recognise the place: the surfaces repeat, the barriers carry graphics, and the light changes. A map also recognises the place best near where its scan began.
- **Immersal recognises the entrance much better.** But an Immersal map is capped at about 500 photos, so the whole route needs several scans. Lining those scans up is the hard part:
 - Lining them up by hand was out by tens of metres.
 - Automatic matching is unreliable on plants.
 - Today's linking walk gave a better fit, but it doesn't yet agree with the marked points.
- **Two phone scans can't be combined.** Each starts from its own origin, and there is no way yet to add to an existing map.
- **Light matters.** Tours will run in the daytime, so scans should be made in daytime light too.
- **Remote testing works.** With Field Test logs and the editor, we could diagnose problems from overseas while our tester was on site.

Next steps

- **On site (Yvonne):** make one Immersal scan of the whole route within the 500-photo limit, split by area (100+ photos at the entrance), ideally in daytime light. If it holds together, we get the whole tour in one map with no lining up needed.
- **App: "Add to an existing map".** The phone first finds itself in the current map, then extends it, so weak areas like the entrance can be scanned again in more detail.
- **App: combined positioning.** Use Immersal to get a position fix, then let the phone's own tracking carry it between fixes.
- **Accuracy:** fix a known iPhone issue in how the phone-side Immersal result is read, then use the linking walk again to settle the five-scan line-up. After that, work on making positioning faster and more accurate.
- **Glasses:** not tested at the Flower Dome yet. Test with the glasses on site once an Immersal map covers the route.